

CO2 - AT2 - Scenario Based Questions[← Back](#)**QUESTIONS****Q1**

Scenario Based - 1. Electric Vehicle Battery Manufacturing System
25 marks

Q2

Scenario Based - 2. Smart Grid Power Distribution System
25 marks

Q3

Scenario Based - 3. A pharmaceutical company manufactured 2,50,000 vaccine doses in 2026
25 marks

Q4

Scenario Based - Airport Passenger Security Screening System
25 marks

4/4 answered

Q4 Scenario Based - Airport Passenger Security Screening System

25 marks

An international airport processes 1,500 passengers daily.

Security Rules:

Age below 18 → Minor Screening

Age 18–59 → Normal Screening

Age 60 and above → Senior Citizen Screening

Carrying baggage above 30 kg → Additional Inspection

Carrying prohibited items → Entry Denied

A) Develop a C program using arithmetic operators to calculate total baggage weight and screening statistics. Use decision-making statements to determine passenger screening status.

B) Use looping constructs to process passengers. Use continue to skip cancelled tickets and break during emergency evacuation.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int i = 1;
5     int age, baggage, prohibited, cancelled, emergency;
6     int total_weight = 0;
7
8     int stop = 0; // flag instead of break
9 }
```

Input

Passenger 1
0
17
20

Output

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Q2 Scenario Based - 2. Smart Grid Power Distribution System

25 marks

A power distribution company monitors 15 substations every hour. Each substation has a maximum capacity of 500 MW.

Business Rules:

Load $\leq 60\%$ $\rightarrow$ Normal

61–85% $\rightarrow$ High Load

Above 85% $\rightarrow$ Critical

Above 500 MW $\rightarrow$ Grid Failure

A) Develop a C program using arithmetic operators to calculate the load percentage and available capacity. Use decision-making statements to classify each substation.

B) Use looping constructs to monitor all substations. Use continue to skip substations under maintenance and break if grid failure occurs.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int i;
5     float load;
6     int max_capacity = 500;
7
8     for(i = 1; i <= 15; i++) {
9         printf("\nEnter load for Substation %d (in MW): ", i);
10        scanf("%f", &load);
```

Input

Substation 1 Load = 100
Substation 2 Load = 310
Substation 3 Load = 430
Substation 4 Load = 600

Output

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Q3 Scenario Based - 3. A pharmaceutical company manufactured 2,50,000 vaccine doses in 2026

25 marks

Vaccines are distributed to 25 hospitals. Each hospital must receive at least 8,000 doses.

Distribution Status:

$\geq 10,000$ doses → Sufficient

8,000–9,999 → Minimum Requirement

Below 8,000 → Emergency Supply Required

A) Develop a C program using arithmetic operators to calculate remaining vaccine stock after distribution. Use decision-making statements to classify each hospital.

B) Use looping constructs to process all 25 hospitals. Use continue for closed hospitals and break if stock becomes zero.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int i = 1;
5     int doses;
6     int total_stock = 250000;
7     int closed;
8     int stop = 0; // flag instead of break
9
10    while(i <= 25 && stop == 0) {
11
12        printf("\nHospital %d\n", i);
```

Input

Hospital 1
0
12000

Output

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25 marks

4/4 answered

Q1 Scenario Based - 1. Electric Vehicle Battery Manufacturing System

25 marks

An electric vehicle battery manufacturing company produced batteries from January 2026 to December 2026. Each month, the target production is 5,000 batteries. If production exceeds the target, the company receives a 10% performance incentive. If production is below 80% of the target, the production unit is marked for inspection.

- A) Develop a C program using arithmetic operators to calculate the annual production, average monthly production, and production efficiency. Use decision-making statements to determine the incentive and inspection status.
- B) Use looping constructs to process production details for all 12 months. Use continue to skip months with invalid production values and break if the plant shuts down due to equipment failure.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int month;
5     int production;
6     int target = 5000;
7     float incentive;
8
9     for(month = 1; month <= 12; month++) {
10         printf("\nEnter production for Month %d: ", month);
11         scanf("%d", &production);
12
13         if(production > target) {
```

Input

Enter production for Month 1: 6000
Is plant shutdown in Month 1? 0

Enter production for Month 2: 3500

Output

